

國立中央大學

光電科學研究所  
碩士論文

半導體雷射泵浦內建式 Q-調制 Nd:MgO:PPLN  
雷射之研究

Diode-pumped internally Q-switched Nd:MgO:PPLN Laser

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中華民國九十五年十月



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## 摘要

體積小高功率的半導體雷射泵浦固態雷射在雷射測距、顯示科技、遠端感應、光儲存、生醫以及海底通訊等方面有許多令人感興趣的應用。而Q調制雷射技術提供一種能重複地生產高能鐳射脈搏的有效方法並且能大大提升雷射的峰值功率，因此對能產生輸出功率大於瓦特等級的微型化固態雷射而言，Q調制雷射已經變成一種重要技術。

另外藉由整合雷射共振腔內元件除了可以減少共振腔對雷射的損耗以及簡化雷射光路的校準，也同時可以達到縮減雷射系統體積的優點。在傳統的雷射系統中，雷射增益物質與其他共振腔內元件(例如Q開關、偏振片、非線性晶體等)是獨立分開的，也因此內建式Q調制雷射與內建式二倍頻雷射在追求系統微小化的過程中提供其他更好的選擇。在本篇論文中，我們率先嘗試在Nd:MgO:LiNbO<sub>3</sub>晶體上整合一塊週期性極化反轉鉍酸鋰晶體(PPLN)來作為電光(EO)式Q開關並加以完成輻射波長為1.085 $\mu$ m的小型半導體雷射泵浦固態雷射。我們成功製作一個Nd:MgO:PPLN並同時整合雷射增益與Q調製的功能。我們的論文中顯示了半導體雷射泵浦內建式Q調制Nd:MgO:PPLN雷射的功能與光學特型藉由一塊長度為12mm、週期為13.6 $\mu$ m的Nd:MgO:PPLN晶體來作為雷射Q開關。當我們對Nd:MgO:PPLN晶體以5kHz的頻率以及施加260V的電壓以及吸收功率為0.61W時來做Q調制時，我們測量到雷射平均功率為7.6mW以及脈衝寬度約30ns，經計算後得知脈衝峰值功率為55W而脈衝能量為1.6 $\mu$ J。本文中已經完成Q調制雷射的製作，以此為基礎加以延伸，再配合PPLN來做波長轉換，相信可以成功的製作出內建式腔內波長轉換元件。

## Abstract

Compact and powerful diode laser-pumped solid-state lasers are of particular interest for many applications, such as laser ranging, display, remote sensing, optical storage, biomedicine, and undersea communication. Laser Q-switching technique has provided an efficient way of producing repetitively high-energy laser pulses to greatly enhance the peak power of the laser and therefore become an important technique in especially a miniature solid-state laser for producing an output power >watt level.

In addition to the merit of system size reduction, monolithic integration of laser intracavity elements has the obvious advantage of reducing the difficulty with resonator loss and laser alignment. While conventional laser systems have laser gain media operated separately to other intracavity elements (e.g., the Q-switches, polarizers, nonlinear crystals, etc.), self-(internal-)Q-switching and self-frequency-doubling lasers have given good alternatives of further system miniaturization. In this work, we demonstrated the first attempt of integrating an electro-optic (EO) PPLN Q-switch in a Nd:MgO:LiNbO<sub>3</sub> crystal to realize a compact diode-pumped internally Q-switched solid-state laser radiating at 1.085 $\mu$ m. We have successfully integrated a 12-mm long, 13.6- $\mu$ m period EO Nd:MgO:PPLN in a 17-mm long Nd:MgO:LiNbO<sub>3</sub> laser gain medium. When the EO Nd:MgO:PPLN Q-switch was driven by a 5 kHz, 260-V voltage pulse train with a 300 ns pulse width, at 0.61-W absorbed pump power, we measured ~7.7 mW average power or 55 W peak power with 1.6- $\mu$ J pulse energy in ~30-ns laser pulse width. Since PPLN is characterized by its flexibility of structure tailoring for phase-matching an arbitrary nonlinear wavelength conversion process, the extension of currently developed internally Q-switched Nd:MgO:PPLN laser to a monolithic all PPLN internally Q-switched intracavity wavelength converter is promising.

## 致謝

在攻讀碩士班的期間，最感謝的是我的指導教授陳彥宏老師對於實驗及論文的指導與建議以及對生活上的照顧，使我能夠順利完成學業。另外感謝清華大學光電所黃衍介老師提供許多實驗製程設備上的幫助並且在口試的過程中給予的指教，另外也感謝口試委員清華大學光電所楊士禮老師以及本所的鍾德元老師能在百忙中抽空擔任口試委員。感謝黃老師實驗室的彥穎學長，碩泰學長以及寵棟學長給予實驗的建議以及許多專業的知識，讓我得以完成實驗製程。在此也由衷的感激昭弘學長對於實驗上的支援以及意見，以及實驗室同學俊育、少偉、韋奴在許多方面的幫忙。感謝學弟兆陞、正良、聖龍、錫雄的完美拋光，讓我能有更多時間撰寫論文。

最後，感謝家裡父母的支持讓我在這段期間內能完成學業，謝謝姐姐在我心情煩悶的時候開解我，家人的關心與照顧，是我能一路走來的原動力。

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## Chapter 1

### Introduction

#### 1.1 Motivation

With the advanced development of lasers, the nonlinear crystals such as lithium niobate ( $\text{LiNbO}_3$ ),  $\text{KH}_2\text{PO}_4$  (KTP),  $\text{NH}_4\text{H}_2\text{PO}_4$  (KDP) and  $\text{LiTaO}_3$  are the common material to be fabricated as optical frequency conversion device like SHG (Second Harmonic Generation,  $\omega_2=2\omega_1$ ), OPG (Optical Parametric Generation,  $\omega_p = \omega_s + \omega_i$ ), SFG (Sum Frequency Generation,  $\omega_1 + \omega_2 = \omega_3$ ) and DFG (Difference Frequency Generation,  $\omega_1 - \omega_2 = \omega_3$ ).  $\text{LiNbO}_3$  is the most powerful, commercial, and popular material among above nonlinear crystals because of its large electro-optical, acoustic-optical and second order nonlinearity coefficients. Periodic poled Lithium Niobate (PPLN) has become a capital device by for the application of quasi-phase-matching (QPM) technique in laser technology recently.

Generally speaking, the wavelength of a laser source depends on the energy level of laser gain medium. If the extraction photon from high energy level is not the interesting wavelength range, it is necessary to take advantage of wavelength conversion device with quasi-phase-matching technique. Recently, more and more laser devices have been invested and implemented for the combination with laser gain medium and electro-optic elements. These kinds of application with nonlinear-optics for QPM method will provide us a variety of the laser devices like self-frequency-doubling laser in  $\text{Nd:LiNbO}_3$  [1] or  $\text{Er:LiNbO}_3$  [2] to investigate green and red light.

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Furthermore, utilizing electro-optic (EO) effect of  $\text{LiNbO}_3$ , an active Q-switch laser [3] was completed by using PPLN as a polarization rotator [4]. Combination with laser gain medium and EO PPLN as a Q-switched laser device is my investigating topic in the thesis by employing the  $\text{Nd:MgO:LiNbO}_3$  crystal.

The photorefractive effect was discovered in lithium niobate crystal by Ashkin *et al.* in 1966 [5]. The photorefractive effect is caused by that inhomogeneous illumination builds up space-charge fields which modulate the refractive index via electro-optic effect. In order to decrease this optical damage, magnesium oxide (MgO) doped in  $\text{LiNbO}_3$  could substantially reduced photorefractive damage caused by the electric field in the material after Zhong *et al* [6] showed that  $\text{LiNbO}_3$  doped with approximately 5% mole or more MgO exhibits a remarkably reduced photorefractive response compare with undope  $\text{LiNbO}_3$ . Applying the technique  $\text{LiNbO}_3$  of co-doped with Nd and MgO, we can successfully employ the interesting property of  $\text{Nd:MgO:LiNbO}_3$  [7] to investigate the optical device as internally-Q-switch laser.

### **1.2 Quasi-Phase-Match (QPM) Technique for the Application of EO PPLN**

The QPM method is firstly proposed by Bloembergen in 1962. It is a useful approach to achieve nonlinear optical frequency conversion process. In the past few years the advancement of PPLN applied QPM method has permitted high-efficiency nonlinear optical frequency conversion. The origin of the attractive characteristics is the periodic nonlinear coefficient caused by the periodic domain structure, thus the reciprocal vector may compensate

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the phase mismatch during the frequency conversion process. However, besides the nonlinear coefficient, other third-rank tensors are also modulated periodically due to the periodic ferroelectric domains in PPLN. Electro-optic (EO) coefficient is one of such tensors. Recently, the EO effect of a PPLN crystal has been investigated [4]. Owing to the phase mismatch,  $\Delta\vec{k} \neq 0$ , the coupling wave between extraordinary wave and ordinary wave can not be established.

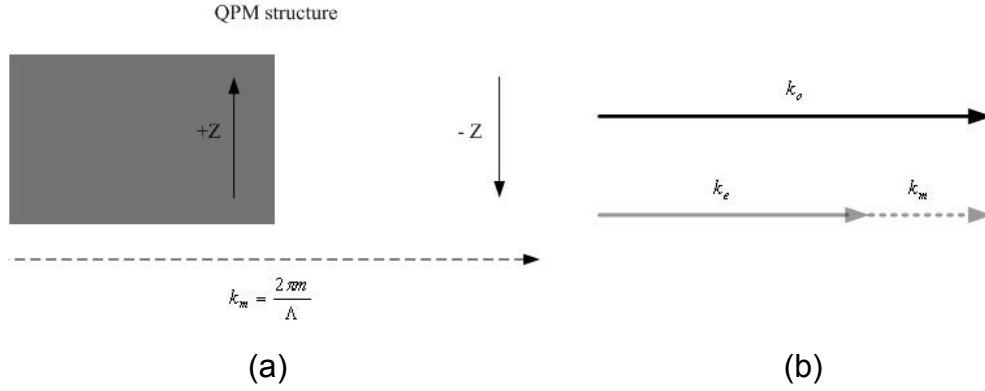
Instead, the coupling wave can be build up between extraordinary wave and ordinary wave in one coherent length and drop in another. The coherence length is defied as

$$L_c = \frac{\pi}{\Delta k} \quad (1.1)$$

It indicates that the destructive interference causes the amplitude to gradually reduce to zero in every  $\pi$  phase difference after the constructive interference. QPM is a technique which provides another grating vector because of the periodic structure to compensate the phase mismatch for the phase velocity dispersion. This compensating term for phase matching  $\Delta\vec{k} = 0$  is given by

$$k_m = k_o - k_e \quad (1.2)$$

Thus the coupling wave can be successfully switched between extraordinary wave and ordinary wave by using QPM technique. Figure 1.1 (a) shows a QPM structure which contains two domain. Along the crystal length, those QPM structure is similar to a grating and the grating period is two times of coherence length ( $\Lambda = 2L_c$ ). As figure 1.1 (b), the QPM structure contributes a grating vector ( $k_m$ ) which compensates the phase mismatch ( $\Delta k$ ) in the phase matching condition.



**Figure 1.1** the QPM structure and wave vector of QPM EO PPLN

### 1.3 Periodic Poled Lithium Niobate (PPLN)

There is a wide transparency range ( $0.35\mu m \sim 5\mu m$ ) and comparably large nonlinear coefficient ( $d_{33} \sim 27 pm/V$ ) in the lithium niobate ( $LiNbO_3$ ) crystal. Here we apply the QPM technique on  $LiNbO_3$  crystal in order to utilize its excellent EO effect. When we make a external electric field alone y-axis, we can exploit the maximum EO coefficient ( $\gamma_{51} \sim 32 pm/V$ ) thus successfully reduce our supplying external voltage. Table 1.1 [8] lists basic properties of  $LiNbO_3$  crystal. Table 1.2 [8] lists electro-optic tensor of  $LiNbO_3$ .

Periodic poled lithium niobate (PPLN) means the polarity (domain) of  $LiNbO_3$  flipped periodically along crystal length for getting QPM condition. For typical ferroelectric materials, electric poling is the most common treatment to fabricate PPLN. It is necessary to achieve the domain inversion by setting the external electric field over the coercive field of  $LiNbO_3$  (21kv/mm). Applying the high voltage power supply, it is much safe and available to complete a PPLN with good duty cycle.

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It is hard to be accomplished in the process of fabrication of Nd:MgO:PPLN due to impurity LiNbO<sub>3</sub>. When LiNbO<sub>3</sub> is doped with some element, the coercive field will change to be less than congruent LiNbO<sub>3</sub> crystal. Taking Er:LiNbO<sub>3</sub> as an example, Yan-Qing Lu *et al.* prove the coercive field reduce to 12 kV/mm in 0.5 mol% Er:LiNbO<sub>3</sub> [9]. Therefore, measuring and using a proper external field for making a Nd:MgO:PPLN device is vital and necessary.

|                                  |                                       |
|----------------------------------|---------------------------------------|
| general properties               |                                       |
| melting point                    | ~1260 °C                              |
| Crystal symmetry                 | rigonal, point group 3 m              |
| method for single crystal growth | Czochralski, large crystals           |
| Hardness                         | ~5 Mohs                               |
| Stoichiometry                    | nonstoichiometric, ~6% Li deficiency  |
|                                  | easy to dope in high concentration    |
| Ferroelectricity                 | Curie temperature TC = 1150 °C        |
|                                  | spontaneous polarization              |
|                                  | Ps(RT) ~78 $\mu$ C/cm <sup>2</sup>    |
| Piezoelectric                    | 180° domains                          |
|                                  |                                       |
| Piroelectric                     |                                       |
| Optical related properties       |                                       |
| transparency region              | VIS/NearIR (~350 nm - ~5 $\mu$ m)     |
| Optical anisotropy               | uniaxial, c-axis                      |
|                                  | no = 2.286, ne = 2.203. (at 632.8 nm) |
| refractive indices               |                                       |
| Optical homogeneity              | $\Delta n \sim 5 \times 10^{-5}$      |

**Table 1.1** Summary of basic properties of LiNbO<sub>3</sub> crystal.[8]

---

electro-optic tensor

---

$$\vec{r}_{ijk} = \begin{bmatrix} 0 & -r_{22} & r_{33} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{bmatrix}$$

Value :  $\gamma_{13} = 10 \text{ pm/V}$ ,  $\gamma_{33} = 32.2 \text{ pm/V}$ ,  $\gamma_{22} = 6.7 \text{ pm/V}$ ,  $\gamma_{51} = 32.6 \text{ pm/V}$

**Table 1.2** Electro-optic tensor of the LiNbO<sub>3</sub> crystal. [8]

## 1.4 Comparison of Laser Gain Medium

There are several common laser gain medium like Nd:glass, Nd:YAG and Nd:YVO<sub>4</sub> for the generation of 1 $\mu\text{m}$ -wavelength laser. As our above mention, we employed Nd:MgO:LiNbO<sub>3</sub> due to its the good performance about EO effect, less photorefractive damage and the Nd<sup>3+</sup> laser ion. Table1.3 shows the difference of the optical characteristics among Nd:YAG, Nd:YVO<sub>4</sub> and Nd:MgO:LiNbO<sub>3</sub>.

| Crystal                 | Nd:YAG(1 at.%)                                                                                              | Nd:YVO <sub>4</sub> (1.1at.%)                                                                               | Nd:MgO:LiNbO <sub>3</sub> (0.2wt%)                                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Emission wavelength     | 946nm,1064nm and 1320nm                                                                                     | 914nm,1064nm and 1340nm                                                                                     | 902nm,1085nm and 1374nm                                                                         |
| Effective cross section | $2.8 \times 10^{-19} \text{ cm}^2 @ 1064 \text{ nm}$<br>$6.4 \times 10^{-20} \text{ cm}^2 @ 946 \text{ nm}$ | $2.5 \times 10^{-18} \text{ cm}^2 @ 1064 \text{ nm}$<br>$2.5 \times 10^{-19} \text{ cm}^2 @ 914 \text{ nm}$ | $1.8 \times 10^{-19} \text{ cm}^2 @ 1085 \text{ nm}$                                            |
| absorption coefficient  | $10 \text{ cm}^{-1} @ 807.5 \text{ nm}$                                                                     | $31.4 \text{ cm}^{-1} (\pi)$ , $10.5 \text{ cm}^{-1} (\sigma) @ 808 \text{ nm}$                             | $1.8 \text{ cm}^{-1} (\pi) @ 814 \text{ nm}$<br>$1.2 \text{ cm}^{-1} (\sigma) @ 808 \text{ nm}$ |
| fluorescence life time  | 230 $\mu\text{s}$                                                                                           | 90 $\mu\text{s}$                                                                                            | 120 $\mu\text{s}$                                                                               |

**Table 1.3** Optical characteristics of Nd:YAG, Nd:YVO<sub>4</sub> and Nd:MgO:LiNbO<sub>3</sub>

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From table 1.3 we can find that Nd:YVO<sub>4</sub> is the best gain medium whatever the quantity of effective cross section and absorption coefficient. Nd:MgO:LiNbO<sub>3</sub> is less than Nd:YVO<sub>4</sub> about 1/10 times but the value of effective cross section is close to Nd:YAG. Although the laser performance of Nd:MgO:LiNbO<sub>3</sub> is not better than Nd:YAG and Nd:YVO<sub>4</sub>, it is easy to implement a miniature laser cavity by the combination with the laser gain medium and the EO crystal.

### **1.5 Dissertation Overview**

In the first chapter, we introduced our motivation to fabricate a simple, compact and miniature Q-switched laser by means of the Nd:MgO:PPLN chip. We wanted to employ the characteristic of a EO PPLN which could rotate the polarization state of the laser beam as the Q switch and integrate the laser gain medium with EO PPLN for a implementation of the internally-Q-switched Nd:MgO:PPLN laser. We also compare the optical property of Nd:YAG and Nd:YVO<sub>4</sub> with Nd:MgO:LiNbO<sub>3</sub>.

In the chapter 2, we presented the phenomenon of EO effect and derived the mode conversion between TE and TM mode under the QPM condition. We also showed that the gain of Nd:MgO:LiNbO<sub>3</sub> depended on the polarization state and therefore Nd:MgO:PPLN was able to be a Q switch due to the characteristic of polarization rotator.

In chapter 3, we showed the process of the fabrication of Nd:MgO:PPLN including lithography, calculation about grating period and the quantity of poling charge for domain inversion and electric poling. After HF etching, we effectively fabricated a Nd:MgO:PPLN.

## **Chapter 1 Introduction**

In chapter 4, we set the diode-pumped internally Q-switched Nd:MgO:PPLN laser and measured the performance of the laser pulse after the fabrication of Nd:MgO:PPLN. We showed the peak power and pulse width of the Q-switched laser as a function of the pumping power and repetition rate. We successfully accomplished Q-switched laser with 28ns pulse width and peak power 55-W. Additionally we measured the relation that pulse width and peak power versus diode pump power and Q-switched repetition rate respectively.

Finally we made a conclusion in the chapter 5 and introduced the prospect based on the Nd:MgO:PPLN .

## Chapter 2

### Theory of the Q-Switched Nd:MgO:PPLN Laser

#### 2.1 The principle of electro-optic effect in LiNbO<sub>3</sub>

The electro-optic (EO) effect [10] is a change of material refractive index under electric field. The refractive index can be madulated by the electric field as a parameter(E). Therefore, the refractive index of a material can be expressby a Taylor expansion

$$n(E) = n + a_1 E + \frac{1}{2} a_2 E^2 + \dots \quad (2.1)$$

Where the  $a_1 E$  term is called the Pockels ffect and the  $\frac{1}{2} a_2 E^2$  term is called the Kerr effect. The Pockels effect is only occurs in a non-central-symmetric material like ADP, KDP, LiNbO<sub>3</sub>, LiTaO<sub>3</sub> and so on.

LiNbO<sub>3</sub> is a negtue uniaxial birefrigence crystal and its extrodinary refractive index is less than ordinary refractive index. Thus, the refractive index is related to the crystal orientation and light polirization. The index ellipsoid can be represented as

$$\frac{x^2}{n_o^2} + \frac{y^2}{n_o^2} + \frac{z^2}{n_e^2} = 1 \quad (2.2)$$

Where the  $n_o$  and  $n_e$  are the ordinary refractive index and extraordinary refractive index respectively. Here we give an impermeability tensor

$$\vec{\eta}_{ij} = \epsilon_0 \vec{\epsilon}^{-1} \quad (2.3)$$

The index ellipsoid is the quadratic representation of the impermeability tensor as

$$\vec{\eta}_{ij} x_i x_j = 1 \quad , \quad i, j, = 1, 2, 3 \quad (2.4)$$

For an EO material, the impermeability tensor will be changed with the variation if electric field and it is given by

$$\Delta \vec{\eta}_{ij} = \vec{r}_{ijk} E_k = \vec{\eta}_{ij}(E_k) - \vec{\eta}_{ij}(0) \quad (2.5)$$

Where  $\vec{\eta}_{ij}(0) = \begin{bmatrix} \frac{1}{n_1^2} & 0 & 0 \\ 0 & \frac{1}{n_2^2} & 0 \\ 0 & 0 & \frac{1}{n_3^2} \end{bmatrix}$

By the above mention from chapter 1, we realize the term  $\vec{r}_{ijk} E_k$  is the EO tensor. The neglect of the second order term and the high order terms, like Kerr effect, is available because its value is too small for the order of the external electric field.  $n_1 = n_2 = n_o$  and  $n_3 = n_e$  can be substitute in eq. (2.5) for LiNbO<sub>3</sub> crystal.

The variation of impermeability tensor is represented as

$$\begin{bmatrix} \Delta \eta_1 \\ \Delta \eta_2 \\ \Delta \eta_3 \\ \Delta \eta_4 \\ \Delta \eta_5 \\ \Delta \eta_6 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix} \quad (2.6)$$

We are able to insert the EO tensor of LiNbO<sub>3</sub> crystal for eq. (2.6)

$$\vec{r}_{ijk} = \begin{bmatrix} 0 & -r_{22} & r_{33} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{bmatrix} \quad (2.7)$$

We can get the 9 impermeability tensor elements based on the parameter of electric field,  $E$ , from eq. (2.6) and eq. (2.7) as a result of applying a external electric field  $\vec{E} = E_y \hat{y}$ .

$$\vec{\Delta\eta}_{ij} = \begin{bmatrix} \Delta\eta_1 & \Delta\eta_6 & \Delta\eta_5 \\ \Delta\eta_6 & \Delta\eta_2 & \Delta\eta_4 \\ \Delta\eta_5 & \Delta\eta_4 & \Delta\eta_3 \end{bmatrix} = \begin{bmatrix} -r_{22}E_y & 0 & 0 \\ 0 & r_{22}E_y & r_{51}E_y \\ 0 & r_{51}E_y & 0 \end{bmatrix} \quad (2.8)$$

It is easily to describe the variation of impermeability tensor from eq. (2.5) and the variation of impermeability tensor is written by

$$\tilde{\eta}_{ij}(E_k) = \begin{bmatrix} \frac{1}{n_o^2} - r_{22}E_y & 0 & 0 \\ 0 & \frac{1}{n_o^2} + r_{22}E_y & r_{51}E_y \\ 0 & r_{51}E_y & \frac{1}{n_e^2} \end{bmatrix} \quad (2.9)$$

We make a substitution into eq.(2.4) by eq.(2.9) and get modified index ellipsoid which can be written by

$$\left( \frac{1}{n_o^2} - r_{22}E_y \right) x^2 + \left( \frac{1}{n_o^2} + r_{22}E_y \right) y^2 + \frac{1}{n_e^2} z^2 + 2r_{51}E_y yz = 1 \quad (2.10)$$

The index variation of the principle axes is represented by the front three terms and the last term is represented that the y-z plane has been rotated by the external electric field  $E_y$ .

After we make the modification of rotation of y-z plane and the neglect of  $r_{22}E_y$  due to the order of  $r_{22}E_y$  comparable to refractive index, we can get

$$\tan 2\theta = - \frac{2r_{51}E_y}{\left( \frac{1}{n_e^2} - \frac{1}{n_o^2} \right)} \quad (2.11)$$

For the assumption of small angle of trigonometric function,  $\theta$  can be written by

$$\theta \approx -\frac{r_{51}E_y}{\left(\frac{1}{n_e^2} - \frac{1}{n_o^2}\right)} \quad (2.12)$$

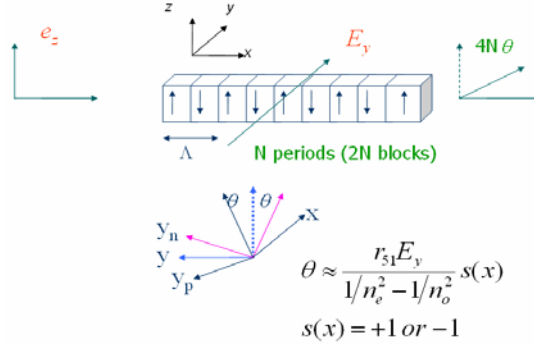
### 2.2 Theory of EO PPLN

EO effect in QPM optical crystal have drawn more and more attentions because of their unique periodic domain inversion structure where the sign of the EO coefficients are modulated. Lu *et al.* [11] reveals an interesting EO PPLN wavelength filter at a birefringence QPM condition where each domain of the periodic structure is equal to the coherence length of optical retardation in LiNbO<sub>3</sub> and thus behaves like a half plate. A PPLN crystal can achieve the same capability as a folded Solc-type optical wavelength filter [11] and successfully control the polarization state of light by utilizing EO effect in the LiNbO<sub>3</sub> crystal.

A PPLN which works as a folded Solc-type optical wavelength filter [12] needs domain period  $\Lambda$  to satisfy

$$\Lambda = m \frac{2\pi}{k_o - k_e} \quad (2.13)$$

where  $k_e$  and  $k_o$  are the wave number of extraordinary (e-ray) and ordinary wave (o-ray) respectively, propagating in the +x direction with odd integer  $m$ . Such a domain inverted structure can rotate a small angle for the crystal axis of the EO PPLN crystal when we use the external electrical field  $E_y$  into the PPLN .



**Figure 2.1** The schematic arrangement of a folded Solc-type filter in PPLN

Figure 2.1 shows the arrangement of a EO PPLN work as a folded Solc-type filter and  $s(x)$  is the sign of EO coefficients. The rotating angle is  $4N\theta$  through each periodic domain inversion structure. As a result, a TM polarization input wave finally transforms to TE state through the whole EO PPLN.

In the presence of an external electric field  $E_y$  alone the PPLN crystallographic  $y$  axis, the dielectric tensor of the crystal is now periodically perturbed in space and can be written as

$$\varepsilon(x, y, z) = \varepsilon_0(y, z) + \Delta\varepsilon(x, y, z) \quad (2.14)$$

Where  $\varepsilon_0(y, z)$  and  $\Delta\varepsilon(x, y, z)$  are the unperturbed part and the periodically varying part along the  $+x$  direction of the dielectric tensor respectively. The electric field of a propagating light with the variation of time in the EO PPLN between e-ray and o-ray can be described as

$$\tilde{E}_{o,e}(x, y, z, t) = \text{Re}[\Phi_{o,e}(y, z)E_{o,e}(x)\exp(i\omega t)] \quad (2.15)$$

Where the  $\Phi(y, z)$  is the electric field profile of transverse mode,  $\omega$  is the angular frequency of the unperturbed mode, and  $E(x)$  is the phasor of the mode, given by

$$E_{o,e}(x) = A_{o,e}(x) \times e^{-jk_{o,e}x} \quad (2.16)$$

Where  $A_{o,e}(x)$  is the field envelope amplitude varying slowly relative to the fast optical carrier frequency. The coupling situation can be described by a set of coupled-mode equation resembling a treatment for a typical codirectional coupling in a folded Solc filter [13].

$$\begin{aligned}\frac{d}{dx}A_o(x) &= -i\kappa A_e(x)e^{i\Delta kx} \\ \frac{d}{dx}A_e(x) &= -i\kappa^* A_o(x)e^{-i\Delta kx}\end{aligned}\quad (2.17)$$

Where the  $\Delta k \equiv (k_o - k_e) - k_m$  is the phase mismatch of e-ray and o-ray coupled in a periodically perturbed structure having the m-th order grating vector  $k_m = \frac{2\pi m}{\Lambda}$  and  $\kappa$  is the coupling coefficient defined by

$$\kappa = \frac{\omega}{4} \gamma_o \gamma_e \int e_o^*(y, z) \cdot \varepsilon_m(y, z) e_e(y, z) dy dz \quad (2.18)$$

Where  $\gamma_{o,e} \equiv \sqrt{\frac{2}{n_{o,e} c \varepsilon_0}}$  is a normalization constant for e-ray and o-ray and the  $n_o$  and  $n_e$  are the ordinary refractive index and extraordinary refractive index respectively with  $c$  the speed of light in vacuum and  $\varepsilon_0$  is the vacuum permittivity,  $e_{o,e}(y, z)$  is the normalized transverse field profile of e-ray and o-ray,  $\varepsilon_m(y, z)$  is the m-th harmonic Fourier expansion coefficient of the periodic perturbation  $\Delta \varepsilon(x, y, z)$ . From eq. (2-15) we can get  $\Phi_{o,e}(y, z) = \gamma_{o,e} e_{o,e}(y, z)$ . Combination with eq. (2-3) and eq. (2-9)

$$\varepsilon = \varepsilon_0 [\eta_{ij}(E_y)]^{-1} \approx \varepsilon_0 n_o^2 n_e^2 \begin{bmatrix} \frac{1}{n_e^2} & 0 & 0 \\ 0 & \frac{1}{n_e^2} & -r_{51} E_y \\ 0 & -r_{51} E_y & \frac{1}{n_o^2} \end{bmatrix} \quad (2.19)$$

From eq. (2.14) and eq. (2.19), we can get

$$\begin{aligned}\Delta\varepsilon(x, y, z) &= -\varepsilon_o \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & r_{51}E_y n_o^2 n_e^2 \\ 0 & r_{51}E_y n_o^2 n_e^2 & 0 \end{bmatrix} g(x) \\ &= \sum_{m \neq 0} \varepsilon_m(y, z) e^{-im \frac{2\pi}{\Lambda} x}\end{aligned}\quad (2.20)$$

The  $g(x)$  is a function periodic in  $x$  with period  $\Lambda$

$$g(x) = \begin{cases} -1 & \text{if } 0 < x < \Lambda/2 \\ +1 & \text{if } \Lambda/2 < x < \Lambda \end{cases} \quad (2.21)$$

It can be expressed by Fourier expansion as

$$\begin{aligned}g(x) &= \sum_{m \neq 0} G_m e^{iK_m x} \\ |G_m| &= \frac{2}{m\pi} \sin(m\pi D)\end{aligned}\quad (2.22)$$

We define a duty cycle  $D \equiv \frac{l}{\Lambda}$  with  $l$  the length of positive domain.

Under the phase matching condition,  $\Delta k = (k_o - k_e) - k_m = 0$ , the propagating wave is coupled perfectly. The grating period is determined by

$$k_m = m \frac{2\pi}{\Lambda} \quad (2.23)$$

Combine eq. (2.22) and eq. (2.23) to eq. (2.20)

$$\varepsilon_m(y, z) = -\varepsilon_o n_o^2 n_e^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & r_{51}E_y \\ 0 & r_{51}E_y & 0 \end{bmatrix} \frac{2}{m\pi} \sin(m\pi D) \quad (2.24)$$

With known  $\varepsilon_m(y, z)$ , the coupling coefficient  $\kappa$  is given by

$$\text{With known } |\kappa| = \frac{\omega}{c} \frac{n_o^2 n_e^2 \gamma_{51} E_y}{\sqrt{n_o n_e}} \frac{\sin(m\pi D)}{m\pi} \quad (2.25)$$

Consider energy conservation within lossless material

$$\frac{d}{dx} (|A_o|^2 + |A_e|^2) = 0 \quad (2.26)$$

Assuming only e-ray (TM polarization) input wave presents so that an initial condition at  $x=0$  is

$$A_o(0) = 0 \quad (2.27)$$

Combination with eq. (2.26) and eq. (2.27) to substitute for eq. (2.17)

$$\begin{aligned} A_o(x) &= e^{i\frac{\Delta\beta}{2}x} \left[ -i\frac{\kappa}{p} \sin px \right] A_e(0) \\ A_e(x) &= e^{-i\frac{\Delta\beta}{2}x} \left[ \cos px + i\frac{\Delta\beta}{2p} \sin px \right] A_e(0) \end{aligned} \quad (2.28)$$

Where  $p^2 = \kappa^* \kappa + (\Delta\beta/2)^2$ . Then the power-conversion efficiency of one polarization mode to the other is give by

$$T(x) = |\kappa|^2 x^2 \sin^2(px) \quad (2.29)$$

When the phase-matching condition is achieved, one can easily derive the voltage required for coupling an initially z-polarized mode (TM mode) to a y-polarized mode at the end-face of the PPLN, given by

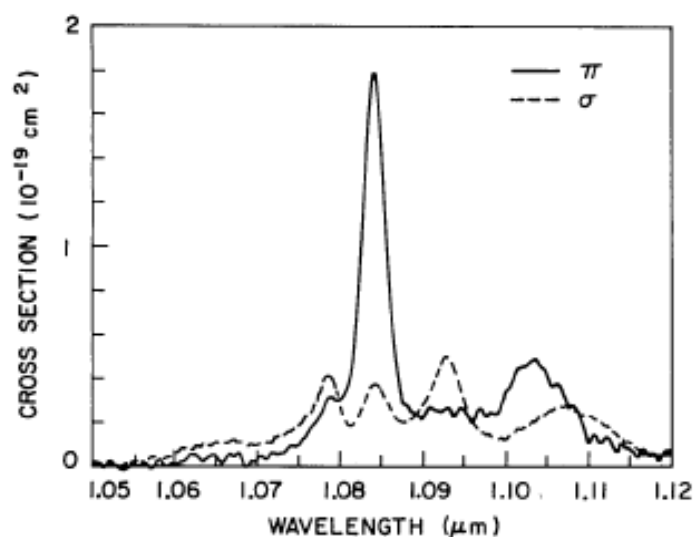
$$V_{90^\circ} = \frac{2q+1}{2} \frac{m\pi}{\sin(m\pi D)} \frac{\lambda}{2} \frac{\sqrt{n_o n_e}}{\gamma_{51} n_o^2 n_e^2} \frac{d}{L_e} \quad (2.30)$$

Where  $q = 0, 1, 2, 3, \dots$  is a positive integer,  $d$  is the eletrode separation in  $y$ , and  $L_e$  is the electrode length in  $x$ .

### 2.3 Laser principle of the Q-Switched Nd:MgO:PPLN Laser

The host material doped with  $\text{Nd}^{3+}$ , such as YAG and  $\text{YVO}_4$ , are the commom gain medium used in a laser system due to dou to the high effective cross section and heat conductivity. But there is no well EO performance in YAG and  $\text{YVO}_4$  crystal. So we choose Nd:MgO:LiNbO<sub>3</sub> as the Q-Switched laser device to make a good combination with a laser gain medium and a EO PPLN which works as a folded Solc-type optical wavelength filter.

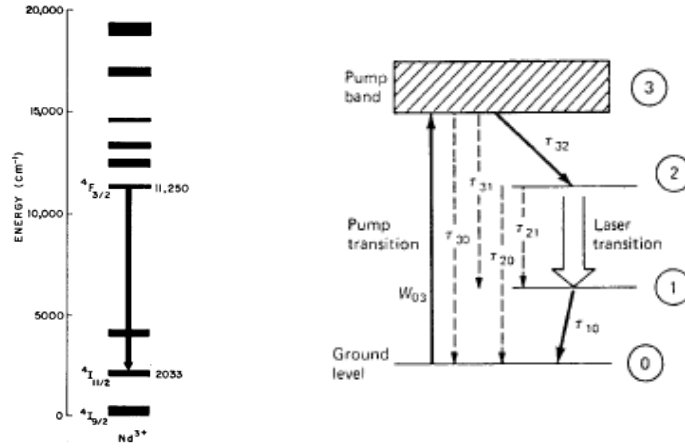
According to spectrum characteristic of Nd:MgO:LiNbO<sub>3</sub> (figure 2.4), Nd:MgO:LiNbO<sub>3</sub> crystal is a polarization-dependent laser gain medium. For the light of  $\pi$ -polarization state, the peak cross section is near  $\sigma_e = 1.8 \times 10^{-19} \text{ cm}^2$  at 1085nm and more than  $\sigma_o = 5.1 \times 10^{-20} \text{ cm}^2$  for  $\sigma$  polarization. When we employ EO PPLN as a intracavity polarization rotator, we can fast switch the laser polarization between high- and low-loss laser modes.



**Figure 2.4** Effective stimulated-emission cross section versus wavelength.[14]

## Chaper 2 Theory of the Q-Switched Nd:MgO:PPLN Laser

It is a four-level system when we use Nd:MgO:LiNbO<sub>3</sub> as the laser gain medium as illustration of figure 2.2 and the population inversion is easily to be achieved comparable to a three-level system.

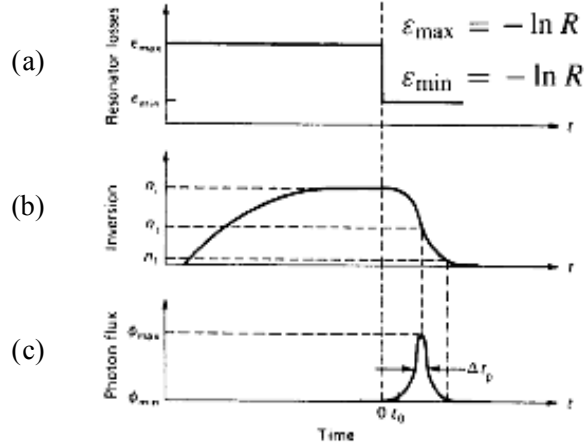


**Figure 2.2** Energy-level diagrams [15] and photon transitions of Nd:MgO: LiNbO<sub>3</sub> [16]

Q-switched laser is one of the laser operations employed for the generation of high pulse power. In the technique of Q-switching, how to make a modulation of tuning the Q value of the resonant cavity is a important problem. The quality factor Q of a laser cavity is defined as eq. (2.28)

$$Q = 2\pi \times \frac{\text{energy stored}}{\text{energy lost per oscillation cycle}} \quad (2.28)$$

The higher the quality factor Q, the lower the resonant loss  $\epsilon$ . When the resonator loss keeps on  $\epsilon_{\max}$  (low Q state) illustrated as figure 2.3(a), population increase gradually until it saturated illustrated as figure 2.3(b). As the stored energy achieve max state, switch fast resonator loss to  $\epsilon_{\min}$  fast (high Q state), and then population decrease immediately to generate a pulse (figure 2.3(c)).



**Figure 2.3** Development of a Q-switched laser about resonant loss, population inversion, and photon flux as a function of time.[16]

To describe the operation of Q-switched laser, we can solve the simultaneous solution of two coupled differential equations for the time rate of the change of the internal photon density in the resonator (eq. 2.29) and the population inversion density  $n$  in the active medium (eq. 2.30) :

$$\frac{\partial \phi}{\partial t} = \phi \left( c \sigma n \frac{1}{L} - \frac{\epsilon}{t_r} \right) \quad (2.29)$$

$$\frac{\partial n}{\partial t} = -\gamma n \phi \sigma \quad (2.30)$$

We generated a pulse and release the photon energy by means of controlling the resonator loss (Q value). The output energy  $E_{out}$  [16] is given from eq. 2.29 and eq. 2.30 :

$$E_{out} = \frac{h \nu A}{2 \sigma \gamma} \ln\left(\frac{1}{R}\right) \ln\left(\frac{n_i}{n_f}\right) \quad (2.31)$$

where  $h \nu$  is the laser photon energy and  $A$  is the effective beam cross sectional area.  $\sigma$  is the stimulated emission cross section, and  $\delta$  is the round-trip loss.  $\gamma$  is one for a four-level system.  $n_i$  and  $n_f$  are the initial and final population inversion density. We can find that  $E_{out}$  is a function of

## Chaper 2 Theory of the Q-Switched Nd:MgO:PPLN Laser

the ratio  $\frac{n_i}{n_f}$  and  $\frac{n_i}{n_f}$  is usually noted  $z$ . The value of  $z$  ( $=\frac{n_i}{n_f}$ ) is depend on cavity loss  $\delta$  (Q state).

So we re-write  $E_{out}$

$$E_{out} = \frac{h\nu A \delta}{2\sigma\gamma} (z - 1 - \ln z) \quad (2.32)$$

The FWHM pulse width versus  $z$  is obtained from

$$t_p = \frac{t_r}{\delta} \left( \frac{\ln z}{z[1 - a(1 - \ln a)]} \right) \quad (2.33)$$

Where  $t_r$  is the cavity round-trip time and  $a = \frac{(z-1)}{z \ln z}$

Thus pulse peak power is obtained from

$$P_{peak} = S_p \frac{E_{out}}{t_p} \quad (2.34)$$

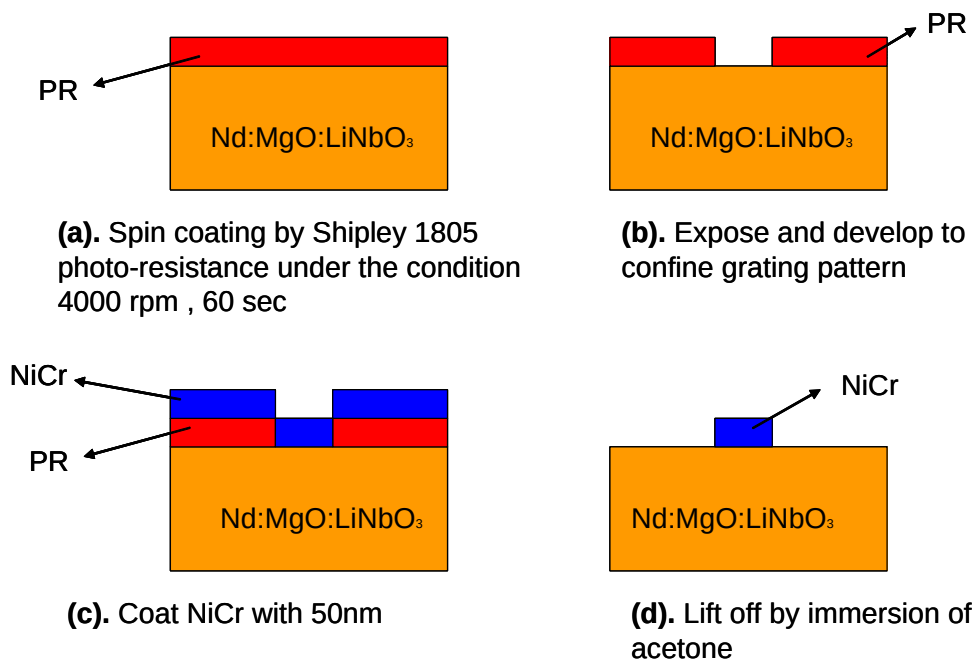
Where  $S_p$  is a number that characterizes the pulse shape.[17]

## Chapter 3

### Fabrication of Nd:MgO:PPLN

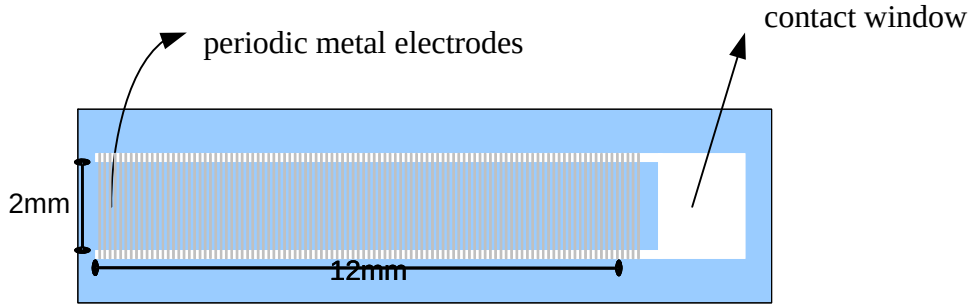
#### 3.1 Electric poling of the fabrication of Nd:MgO:PPLN

A z-cut 0.4mol.% Nd and 5mol.% MgO-codoped LiNbO<sub>3</sub> with 1-mm thickness was used. As we mentioned to produce the periodical structure on the Nd:MgO:LiNbO<sub>3</sub> wafer in the chapter 2, the periodically external electric field was the the most effective method to fabricate our Nd:MgO:PPLN. Electecic poling is one of the common types of the fabrication of PPLN. In this paper, we do the lithographic process on the +z surface of a Nd:MgO:LiNbO<sub>3</sub> wafer to define the grating pattern and a contact window to touch the electrode with high-voltage (Figure 3.3 (a) and (b)). Then we coated Nd:MgO:LiNbO<sub>3</sub> with NiCr about 50-nm-thick as the metal electrode by a RF sputter evaporater (Figure 3.3 (c)). Finally immersed the Nd:MgO:LiNbO<sub>3</sub> sample in acetone organic solvent to do lift-off process (Figure 3.3 (d)).



**Figure 3.1** The schematic procedure of the lithographic process for Nd:MgO:PPLN

We make the periodic metal electrode to complete electric poling for the Nd:MgO:PPLN. The poling pattern include two parts. One is the periodic domain inversed area and the other is contact window to connect with the perioic domain inversed part.



**Figure 3.2** The configuration of the poling pattern

The grating period determined under QPM condition mentioned in chapter 1 and chapter 2 is given by

$$\Lambda = 2L_c = \frac{2\pi}{\Delta k} = \frac{2\pi}{k_o - k_e} = \frac{\lambda}{n_o - n_e} \quad (3.1)$$

Acorrding to reference [18] published by G.K. Kitaeva *et al.* in 1998, the refractive index of Nd:MgO:LiNbO<sub>3</sub> could be calculated by the modified Sellmeier eq. and got  $n_o = 2.2288$  and  $n_e = 2.1485$  respectively. Thus we were able to get grating period  $\Lambda \sim 13.6 \mu\text{m}$  at room temperate.



**(a).** grating area

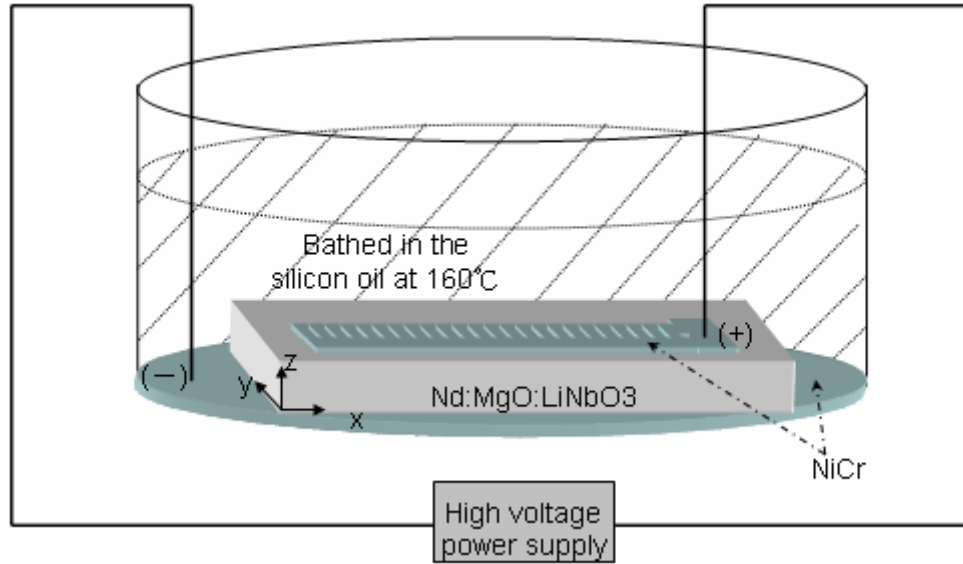


**(b).** contact window

**Figure 3.3** The result of lithography process of poling electrodes

### Chapter 3 Fabrication of Nd:MgO:PPLN

After lithography process of fabrication of the electrodes for electric field poling (Figure 3.2), the wafer was immersed in the hot silicon oil around 160 °C and prepare to do the electric poling process. The setup was illustrated as figure 3.3 .

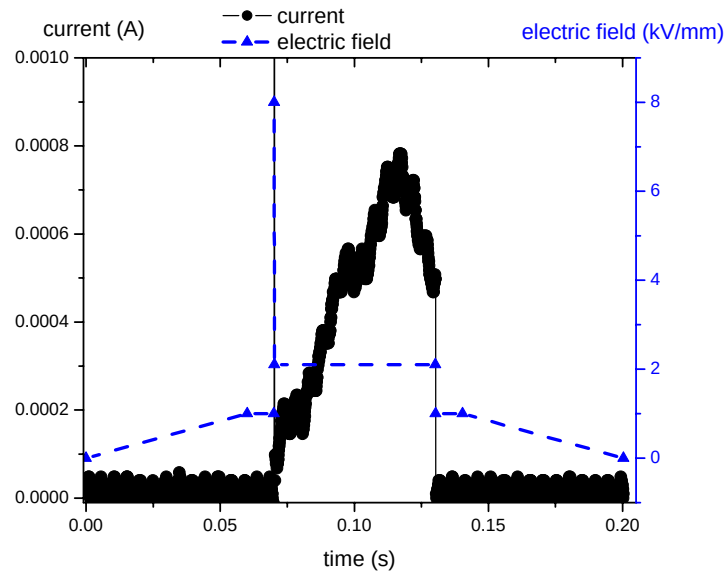


**Figure 3.4** The configuration of electric field poling technique for fabricating Nd:MgO:PPLN

Some domain kinetics including domain nucleation at the electrode edge , domain tip propagation toward the opposite face of the crystal, termination of the tip at the opposite side of the crystal, rapid coalescence under the electrodes, sideways propagation of the domain wall, and the stabilization of the new domains are happened during the electric field poling [19]. After realizing these phenomena of electric field poling process, we have to compute and calculate the proper quantity of electric charge to completely fabricate a Nd:MgO:PPLN crystal with good duty quality about 50%.

$$Q = 2P_s \times \{[(\text{area of grating}) \times 0.5] + [\text{area of electrode}]\} \quad (3.1)$$

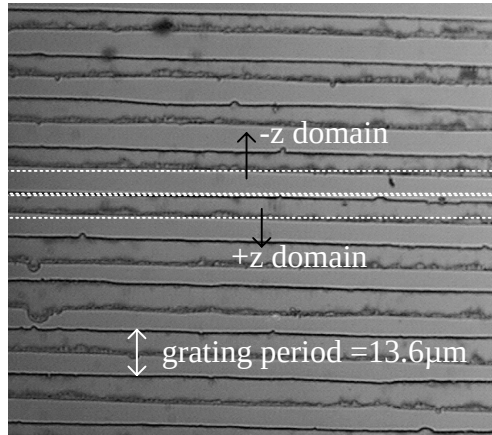
The constant,  $P_s$ , is the spontaneous polarization density of Nd:MgO:LiNbO<sub>3</sub> and the value is  $80 \mu\text{C}/\text{cm}^2$  [20]. When the grating pattern was designed about the area of  $0.2 \text{ cm} \times 1.2 \text{ cm}$  and additionally the area of electrodes is included, the suitable quantity of charge was  $31.52 \mu\text{C}$ . After HF etching process, it was easy to observe the result of the electric poling upon the +z or -z surface due to the faster etching rate on -z domain than that on the +z domain.



**Figure 3.5** The electric field and current versus time

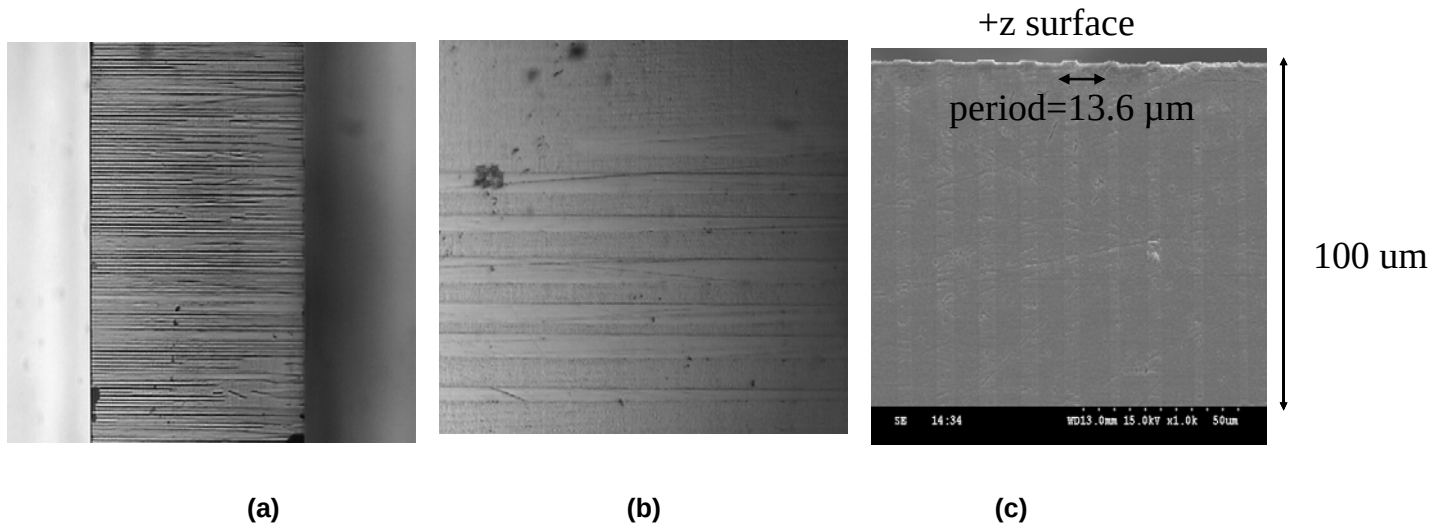
We set the poling field  $2.1 \text{ kV/mm}$  at  $160^\circ\text{C}$ . Figure 3.5 showed the electric field and current versus time respectively. Under the condition illustrated as figure 3.5, the poling charge was about  $28.7 \mu\text{C}$  and less than theoretical value we calculated.

Figure 3.6 is the situation after HF etching on the +z surface and shows the poling result of Nd:MgO:PPLN. It appears that periodic domain inversed structure is formed and the duty cycle is achieved 40%.



**Figure 3.6** HF etching on the +z surface

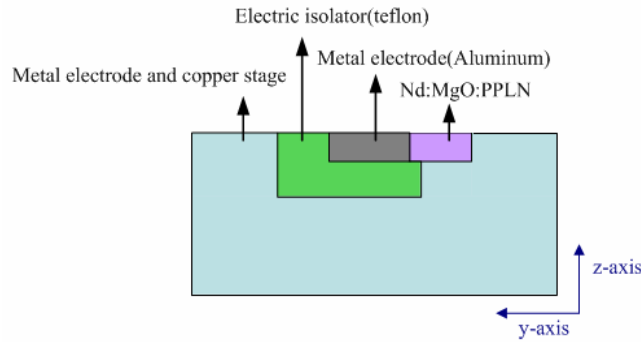
After observation of the +z surface, we diced the Nd:MgO:PPLN chip in the middle and then see the y-cut surface to ensure that the periodic domain inversion parts have been penetrated through the Nd:MgO:LiNbO<sub>3</sub> from the +z surface to the -z surface. Figure 3.7 reveal that the Nd:MgO:PPLN is formed and the domain wall is straight. The pictures also exhibit that the domain is inversed periodically through the 1mm-thick Nd:MgO:LiNbO<sub>3</sub> from the +z surface to the -z surface.



**Figure 3.7** HF etching on the y-cut surface

#### 3.2 Experimental setup of EO Nd:MgO:PPLN

In order to utilize EO effect of  $\text{LiNbO}_3$  crystal and the largest electrooptic coefficient  $\gamma_{51}$ , it is necessary to apply an external electric field along the y-axis. How to construct a set of parallel electrode for the Nd:MgO:PPLN to make a uniformly external electric field is a vital problem. Additionally, Nd:MgO:PPLN is also a laser gain medium in the experimental setup. Thus we need a copper stage to set a thermal-electric (TE) cooler and a water-cooled copper housing to dissipate the excess heat from the Nd:MgO:PPLN chip as laser gain medium and tune the temperature in Nd:MgO:PPLN chip for achieving phase-matching condition.



**Figure 3.8** The transverse structure of the experimental arrangement of EO Nd:MgO:PPLN

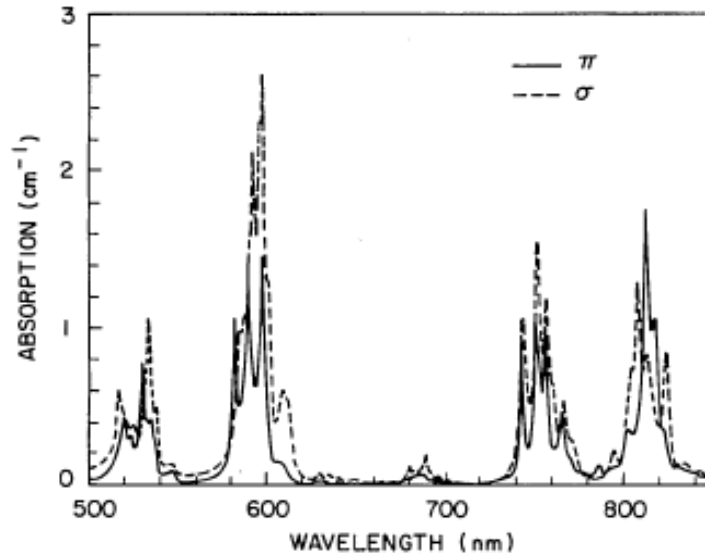
After polishing the both sides of end faces and accomplishing AR coating ( $\text{AR} = 99.7\% @ 1085\text{nm}$  &  $\text{AR} = 97\% @ 808\text{nm}$ ) on the end faces, we used the dicing machine to cut the Nd:MgO:PPLN as short as 1mm-length for the purpose of allowing application of an external voltage. Then we smeared electric conductive silver inks on the both sides of x-z cross faces directly to make the Nd:MgO:PPLN more compact with the both sides of metal electrodes and mounted the chip on the stage and illustrated as Figure 3.7.

## Chapter 4

### Experimental analysis

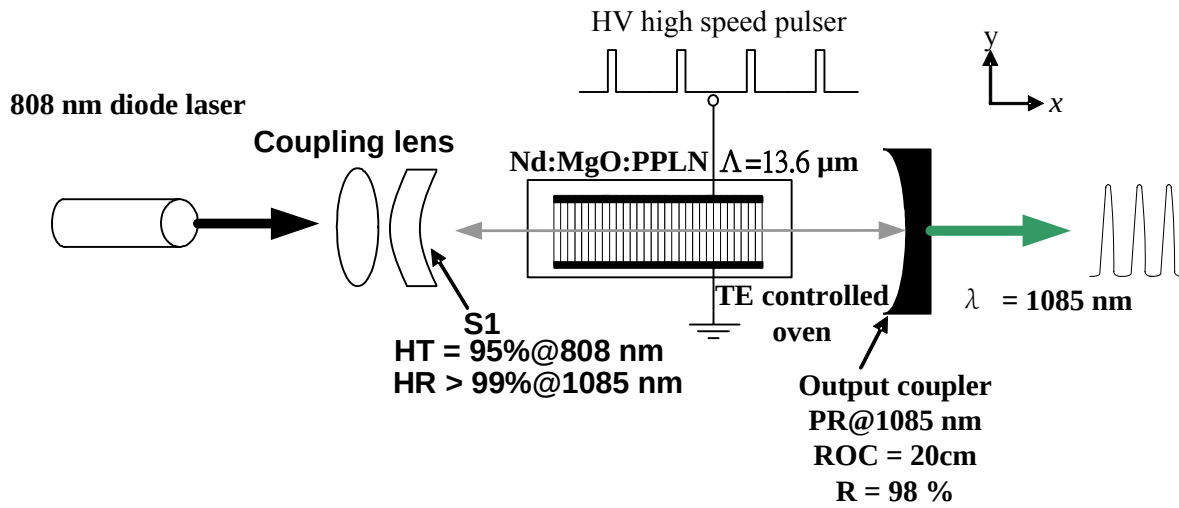
#### 4.1 Experimental setup

We fabricated a 17-mm-length (including 5-mm-length gain medium), and 1-mm-thick Nd:MgO:PPLN with grating period  $\Lambda = 13.6 \mu\text{m}$  to implement the internally Q-switched laser. We also carried out the mechanical stage to generate the electric field along the y-direction by applying the DC voltage. According to the absorption spectrum of Nd:MgO:LiNbO<sub>3</sub> [21], we selected 808 nm diode laser as the pumping source. We can figure out that the  $\sigma$ -pol. light is a suitable polarization state for 808 nm diode laser due to its central wavelength is close to 808 nm and higher absorption coefficient than  $\pi$ -pol from figure 4.1.



**Figure 4.1** The absorption spectrum of Nd:MgO: LiNbO<sub>3</sub> [1]

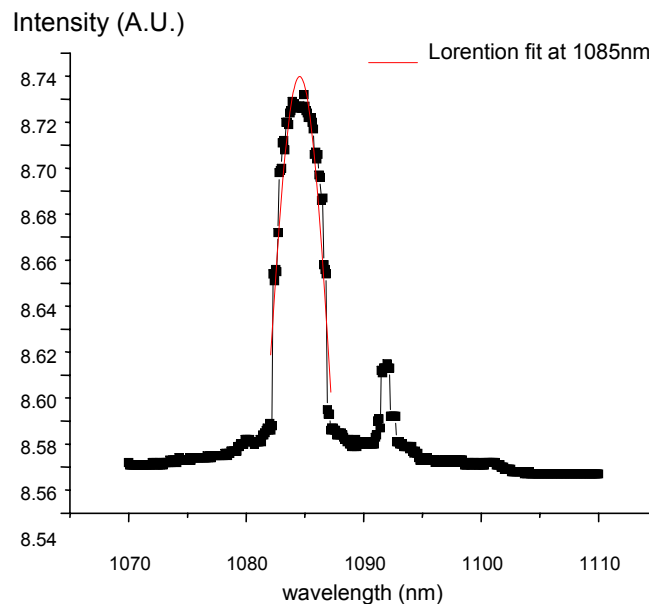
Figure 4.2 shows the schematic of internally-Q-switched Nd:MgO:PPLN laser. The both sides of the Nd:MgO:PPLN were coated with anti-reflection (AR) coating at 808 nm and 1085 nm. The output coupler was a plano-concave mirror with 98% reflectance at 1085. The laser cavity length was approximately 4cm.



**Figure 4.2** The schematic of diode-pump internally-Q-switched Nd:MgO:PPLN laser

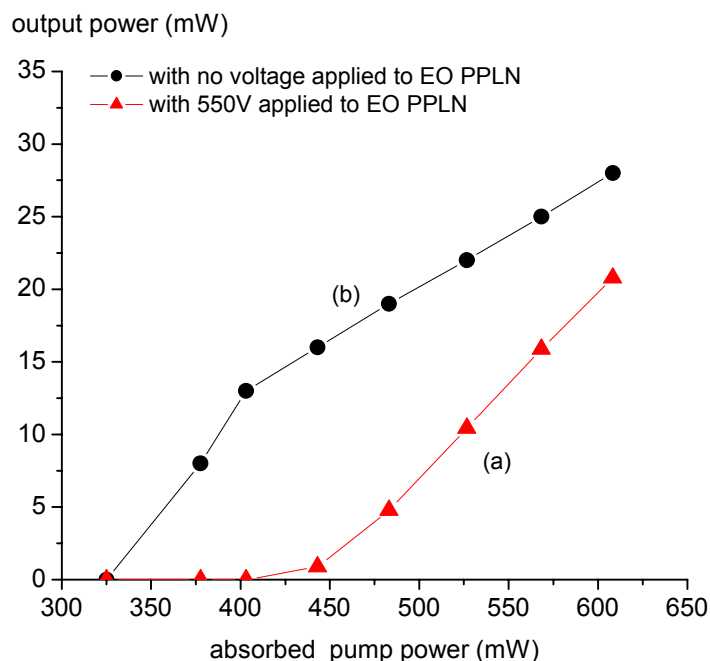
The Nd:MgO:PPLN was mounted in a TE controlled oven which could tune the temperature between  $0^\circ\text{C} \sim 100^\circ\text{C}$ . The large range of the temperature is convenient for our operation to achieve phase-matching condition and get the best conversion efficiency of switching polarization mode. We controlled the temperature at the  $42^\circ\text{C}$  for phase-matching. If we wanted to operate at room temperature, the grating period should deduce to  $13.4 \mu\text{m}$  or  $13.5 \mu\text{m}$ .

Figure 4.3 shows the output spectrum of the Nd:MgO:PPLN laser system .



**Figure 4.3** The CW output wavelength of the Nd:MgO:PPLN laser system

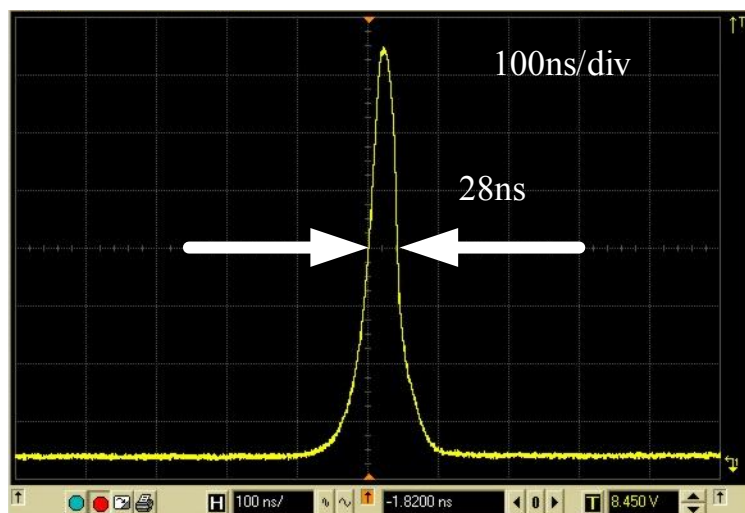
The continuous-wave (CW) performance of the Nd:MgO:PPLN laser system is plotted in figure. 4.4, where curve (a) corresponding to a 50-C voltage in the EO PPLN crystal and curve (b) corresponding to co voltage in the EO PPLN crystal.



**Figure 4.4** The CW output power of the Nd:MgO:PPLN laser system as a function of input absorbed pump power with (a) DC 550-V and with (b) no voltage applied to EO PPLN crystal.

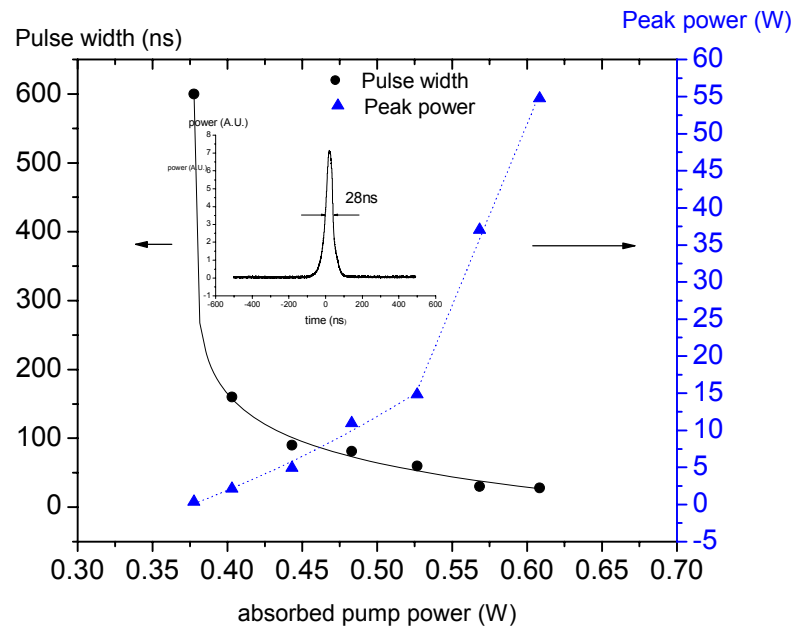
### 4.2 Experimental result

At a 260-V Q-switched voltage, we measured a 28-ns, 1.6- $\mu$ J Q-switched pulse, as shown in figure 4.5, corresponding to a n average output power of 7.6-mW at 5-kHz repetition rate.

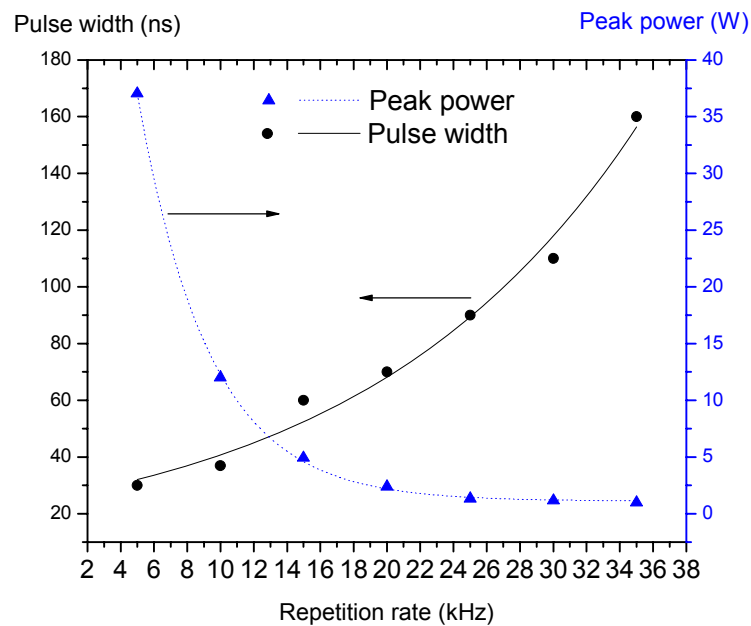


**Figure 4.5** The measured minimum Q-switched pulse from the internally-Q-switched Nd:MgO:PPLN laser at 0.61-W absorbed power and 5 kHz repetition rate. The minimum pulse width was 28 ns and the pulse energy was about 1.6  $\mu\text{J}$ .

Additionally, we keep the operational temperature on  $42^{\circ}\text{C}$  and tune the diode pump power and repetition rate to observe the change of pulse width and peak power. Firstly we increased the absorbed pump power from 0.38-W to 0.61-W and the pulse width decreased from  $\sim 600$  to  $\sim 30$ -ns. According to the eq. (2.34) we calculated the peak power for each Q-switched pulse. The peak power was changed from 0.38-W to 55- W. The data were shown as figure 4.6. On the other hand, we tuned the Q-switch repetition rate and measure the pulse width and peak power with ab 0.57-W sorbed pump power, shown as figure 4.7.



**Figure 4.6** The measured pulse width and peak power versus the absorbed pump power.



**Figure 4.7** The measured pulse width and peak power versus the Q-switch repetition rate at 0.58-W absorbed pump power.

For the CW pumped Q-switched laser, the pulse width was determined according to the model in Ref. [22,23] , given by

$$p_w = p_s \frac{\gamma \eta(\gamma)}{\gamma - 1 - \ln \gamma} \tau_c \quad (4.1)$$

with the energy-extraction efficiency  $\eta = 1 - e^{-\gamma}$ , where  $p_s$  is the pulse shape function,  $\tau_c$  is the cavity photon life time, and  $\gamma$  is the initial population-inversion ratio just after Q-switching.  $\gamma$  is a function of CW pumping rate, laser upper level life time and Q-switching repetition rate.

With the Ref. [22,24], the peak power was a function of Q-switch repetition rate and given by

$$P_{peak} = \frac{-\ln R}{8} \frac{\pi \omega_0^2}{\tau_c \sigma} h \nu (\gamma - 1 - \ln \gamma) \quad (4.2)$$

where  $h$  is Planck's constant,  $\nu$  is the laser frequency,  $R$  is the output coupler reflectance (0.98 in our case),  $\sigma = 1.8 \times 10^{-19} \text{ cm}^2$  is the stimulated cross section of the Nd:MgO:LiNbO<sub>3</sub> crystal and the  $w_0$  is the laser radius.

From eq. 4.1 and eq. 4.2 , we know that pulse width  $p_w$  and peak power  $P_{peak}$  function of population inversion ratio  $\gamma$  . In Ref.[22] ,  $\gamma$  is given by

$$\gamma_i^{(k+1)} = \gamma_{cw} - (\gamma_{cw} - \gamma_f^{(k)}) e^{-\frac{\gamma_2}{f_r}} \quad (4.3)$$

$\gamma_i^{(k+1)}$  and  $\gamma_f^k$  mean the initial population ratio of the  $k+1$ -th and  $k$ -th pulse under the repetitive laser Q-switching condition with CW pumping.  $\gamma_{cw}$  is the normalized CW pumping rate and  $\gamma_2$  is the population decay rate.  $f_r$  is Q-switching repetition rate. When we increase diode pump power ( $\gamma_{cw}$  increasing), we get higher  $\gamma$  Thus  $p_w$  become less and  $P_{peak}$  become more. When we increase Q-switching repetition rate ( $f_r$  increasing), on the other hand,  $\gamma$  decrease and thus  $p_w$  become more and  $P_{peak}$  become less.

## Chapter 5

### Conclusion

#### 5.1 Conclusion and the contribution of the thesis

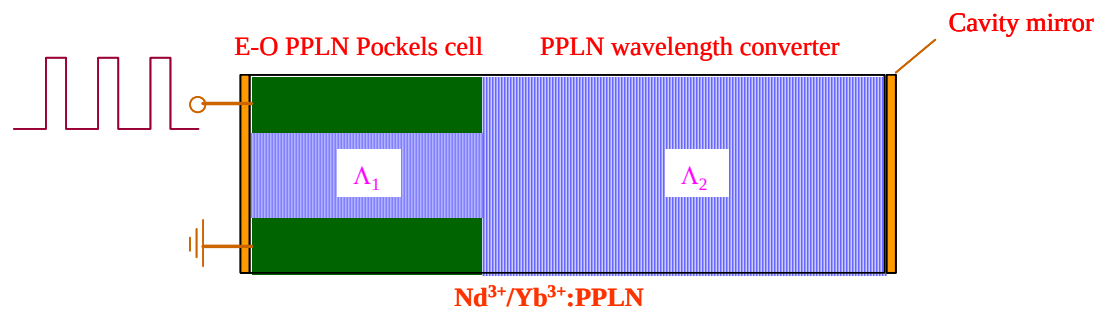
In this dissertation, we concentrated on the application of electro-optic (EO) effect in the Nd:MgO:LiNbO<sub>3</sub> crystal and the integration of the laser gain medium and EO material. We have successfully demonstrated a novel active Q-switched laser by employing Nd:MgO:PPLN as the gain medium and laser Q-switch.

We obtain 40-ns, 45-W Q-switched laser pulses with 0.61-W absorbed pump power when we driving the Nd:MgO:PPLN polarization rotator at 260-V and 300-ns voltage pulse at 5-kHz repetition rate.

Many researches about the investigation of nonlinear devices to prevent from photorefractive damage by using MgO doped in LiNbO<sub>3</sub> were reported. Quasi-phase-match (QPM) technique is an effective application of nonlinear devices. In the thesis, we accomplished the fabrication of Nd:MgO:PPLN in the silicon oil at 160°C. But the domain inversion on the  $-z$  surface is not uniform due to the variation of electric field on the  $-z$  surface. The phenomenon resulted in the poling quality of Nd:MgO:PPLN and led to the high Q-switching voltage.

### 5.1 Prospect

There is also well performance of the wavelength conversion due to high nonlinear coefficient in  $\text{LiNbO}_3$ . In the thesis, we successfully completed a diode-pumped internally Q-switched  $\text{Nd:MgO:PPLN}$  laser by means of EO PPLN as a polarization rotator. We advancedly design other part of PPLN for wavelength conversion to implement a cascaded with EO PPLN as we mentioned. If a PPLN Q-switch is monolithically cascaded with a wavelength converter, such as Second Harmonic Generation (SHG) or optical parametric oscillator (OPO), a highly efficient and integrated nanosecond intracavity SHG or OPO [1] was demonstrated respectively.



**Figure 5.1** Monolithic All-PPLN internally Q-switched and self-frequency-conversion lasers

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